A decade of digital transition, educational equity and adolescent well-being: Evidence from PISA in Viet Nam and East/Southeast Asia, 2012–2022

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Abstract

Between 2012 and 2022, home internet access among Vietnamese 15-year-olds rose from 29% to 90%, the fastest rise among ten East and Southeast Asian PISA systems. We combine four PISA cycles (250,587 students), school questionnaires, population indicators and phone-policy records to examine concurrent changes in well-being and achievement. Viet Nam retained paper-based assessment throughout. Mathematics anxiety met scalar invariance and increased by 0.33 SD; reported loneliness rose from 6.1% to 14.4%. Belonging failed scalar invariance: freeing all three negative-item intercepts attenuated its estimated latent decline from −0.250 to −0.035 SD, while partial invariance still failed. Item-specific exclusion trends should therefore not be read as a validated decline in general belonging. Five reweighting specifications did not assign the loneliness increase to measured home access, but cannot exclude unmeasured exposure or confounding. In 2022, the adjusted home-computer mathematics association was approximately 23 points; no cross-cycle growth in this association is inferred. In 2022, the distraction–achievement association attenuated after conditioning on non-digital disorder; the distraction–anxiety association persisted in Viet Nam after multiple-testing correction. Phone bans were associated with less distraction (−0.08 SD), without statistically clear adjusted achievement or anxiety differences. School-grouped validation added regression calibration and a separate below-Level-2 classifier (Brier 0.164 versus 0.203 at baseline). These observational patterns are consistent with attention to school climate, connectedness and equity; they do not identify causal effects of digital access or phone policies.

Keywords: digital transition, adolescent well-being, loneliness, mathematics anxiety, digital distraction, PISA, measurement invariance, school phone policy

Tóm tắt. Từ 2012 đến 2022, tỷ lệ học sinh 15 tuổi ở Việt Nam có internet tại nhà tăng từ 29% lên 90%, nhanh nhất trong 10 hệ thống giáo dục Đông Á và Đông Nam Á được khảo sát. Nghiên cứu kết hợp bốn chu kỳ PISA (250.587 học sinh), bảng hỏi nhà trường, chỉ số dân số và hồ sơ chính sách điện thoại. Lo âu toán học đạt bất biến đo lường cấp scalar và tăng 0,33 độ lệch chuẩn; tỷ lệ báo cáo cô đơn tăng từ 6,1% lên 14,4%. Tuy nhiên, thang đo thuộc về

không đạt bất biến scalar: giải phóng intercept của cả ba mục phủ định làm mức giảm tiềm ẩn ước tính thu hẹp từ $-0,250$ xuống $-0,035$ độ lệch chuẩn, và mô hình partial-scalar vẫn không đạt tiêu chí bất biến. Vì vậy, xu hướng từng mục về cảm giác bị loại trừ không chứng minh mức suy giảm chung của cảm giác thuộc về. Năm cách tái trọng số không quy mức tăng cô đơn cho tiếp cận thiết bị tại nhà, nhưng không loại trừ tác động của hành vi sử dụng hay yếu tố gây nhiễu chưa đo. Năm 2022, chênh lệch điểm toán gắn với máy tính tại nhà khoảng 23 điểm; nghiên cứu không suy ra mức tăng của chênh lệch này qua chu kỳ. Liên hệ giữa xao nhãng số và điểm toán thu hẹp khi kiểm soát mất trật tự phi kỹ thuật số; liên hệ với lo âu vẫn tồn tại ở Việt Nam sau hiệu chỉnh kiểm định bội. Cấm điện thoại gắn với ít xao nhãng hơn, nhưng chênh lệch điểm toán và lo âu sau điều chỉnh chưa rõ về mặt thống kê. Kiểm định chéo theo trường bổ sung đồ thị hiệu chuẩn và bộ phân loại riêng cho toán dưới mức 2 (Brier 0,164 so với 0,203 ở mô hình mốc). Đây là các mối liên hệ quan sát, không phải ước lượng tác động nhân quả của thiết bị hay chính sách.

1. Introduction

Viet Nam's education system went online in a decade. Internet use in the population rose from 37% in 2012 to 79% in 2022 (World Bank, n.d.). National programmes such as *Sóng và máy tính cho em* ("Connectivity and computers for students") and pandemic-era remote schooling brought devices into most homes (UNESCO Global Education Monitoring Report, n.d.). Vietnamese 15-year-olds have long outperformed what the country's income would predict in PISA (Dang et al., 2020). Whether their well-being has kept pace with this transition is unknown. The question matters beyond Viet Nam: middle-income education systems across Asia are following the same path and are being urged to build "smart, connected" education ecosystems.

International research is divided on what digital transitions do to adolescents. One strand documents rising adolescent loneliness and falling achievement from around 2012 and links both to the spread of smartphones, largely through *aggregate* trends (Twenge, 2026; Twenge et al., 2021). A second strand finds that *individual-level* associations between technology use and well-being are small (Orben & Przybylski, 2019; Przybylski & Weinstein, 2017) and have not grown over time (Vuorre et al., 2021). The two positions are not necessarily contradictory. Societal digitalisation could change the social ecology of adolescence, with consequences for every student, even if a given student's own access matters little. Distinguishing these *ecological* and *individual-exposure* accounts requires data that combine trends with within-cohort associations, which most studies lack.

Classroom-level evidence is similarly mixed. Phone use during lessons impairs learning in experiments (Kuznekoff & Titsworth, 2013; Mendoza et al., 2018), and phone bans improved

low achievers' scores in England (Beland & Murphy, 2016), but the "brain drain" effect of a phone's mere presence finds little meta-analytic support (Böttger et al., 2023; Ward et al., 2017). Policy has moved faster than evidence: a growing share of systems now restrict phones in schools (UNESCO, 2023). Viet Nam permits phones in class for learning purposes with the teacher's permission (Ministry of Education and Training of Viet Nam, 2020).

This study contributes in four ways. **First**, it provides a decade-long account with measurement tests of Vietnamese adolescents' well-being in the digital transition. It uses four PISA cycles in which, uniquely in the region, Viet Nam's administration mode never changed. **Second**, it tests the individual-exposure account directly, through within-cycle associations and a reweighting decomposition of the trend. **Third**, it separates digital distraction from general classroom disorder and links both to achievement and achievement emotions. **Fourth**, it combines student, school, system and policy data. School policies are evaluated with propensity-score weighting, and inference across 13 systems is corrected for multiple testing.

2. Theoretical framework and hypotheses

Displacement and the ecology of adolescence. The displacement hypothesis holds that time online crowds out face-to-face interaction and thereby erodes social connectedness (Kraut et al., 1998). Applied to individuals, it predicts that students with more digital access report more loneliness. Applied to cohorts, it suggests that when connectivity becomes universal, the social environment of school itself changes. Peers are less available and attention is fragmented, which affects connected and less-connected students alike (Twenge et al., 2021). The "Goldilocks" and specification-curve literature, by contrast, suggests that individual-level associations are small and non-linear (Orben & Przybylski, 2019; Przybylski & Weinstein, 2017). Belonging is a fundamental need (Baumeister & Leary, 1995) and a robust correlate of school functioning (Allen et al., 2018), so changes in it are consequential either way.

Attention, disorder and achievement emotions. Digital distraction competes for the limited working memory needed for mathematics (Sweller, 1988), but it also co-occurs with other forms of classroom disorder. Control-value theory (Pekrun, 2006) predicts that a classroom in which students cannot control their attention lowers perceived control over learning and therefore increases anxiety, even when the loss of instruction time is accounted for.

Self-regulation and school policy. Self-regulated learners manage their environment to protect attention (Zimmerman, 2002). Turning off notifications is a concrete instance of this. School phone bans remove the stimulus rather than build the skill, and schools may adopt them in response to existing problems.

Research question and hypotheses. **RQ1 (descriptive)**: How did responses to school-connectedness items and measurement-tested mathematics anxiety differ between Vietnamese PISA cohorts from 2012 to 2022? **H2 (individual exposure)**: within cycles, students with more home digital access report more loneliness and anxiety, and changes in access account for the trend. **H3**: digital distraction is associated with lower achievement and attenuates after conditioning on general classroom disorder. **H4**: digital distraction is associated with higher mathematics anxiety beyond classroom disorder and relationships. **H5**: school phone bans are associated with less distraction; digital self-regulation is associated with higher achievement. Figure 1 summarises the framework.

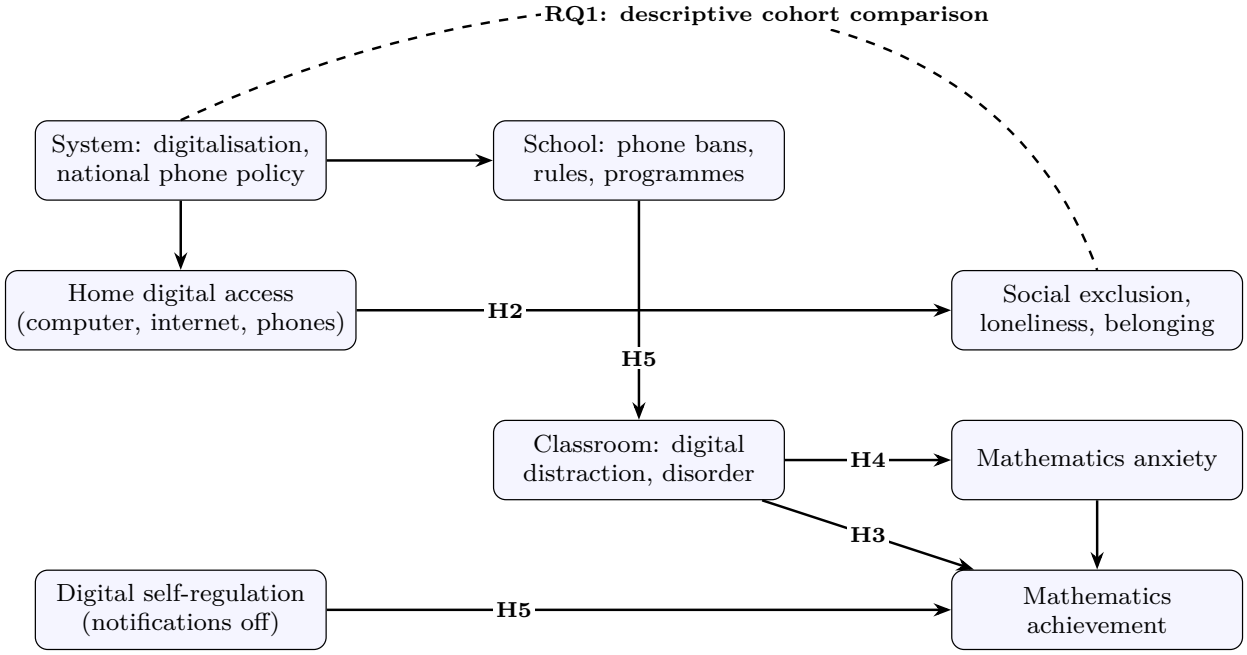


Figure 1: Conceptual framework and hypotheses. The dashed connector locates the descriptive cohort comparison (RQ1); it is not a tested ecological mechanism. Solid arrows from home access represent the individual-exposure hypothesis (H2). Arrows are hypothesised relationships, not identified causal effects.

3. Data

Table 1 lists the five data sources. All are public and downloadable without registration.

Table 1: Data sources and levels of analysis

#	Source	Level	Content and use
S1	PISA 2012, 2015, 2018, 2022 student files (OECD, 2023a)	Student	250,587 students in 10 systems for trends (Viet Nam: 4,959; 5,826; 5,377; 6,068) and 88,087 students in 13 systems in 2022. Harmonised items on digital access, belonging, mathematics anxiety and classroom climate; digital distraction and digital habits (2022).
S2	PISA 2022 school questionnaire	School	Principal reports on phone bans, rules on device use, responsible-internet programmes, SEL policy, school size, type and location.
S3	World Bank WDI (World Bank, n.d.)	System	Internet users, mobile subscriptions, GDP per capita, 2010–2023.
S4	WHO Global Health Observatory (World Health Organization, n.d.)	System	Suicide mortality (SDG 3.4.2), 2000–2021, used as population mental-health context.
S5	UNESCO GEM PEER profiles (UNESCO Global Education Monitoring Report, n.d.)	Policy	National rules on phones in schools, coded with verbatim evidence.

Harmonisation. We use items with identical wording and response scales across cycles: computer for school work and internet at home (in 2022 the internet item excludes access through smartphones); three or more internet-enabled phones at home (2015–2022; 2022 categories mapped to the same threshold); the six sense-of-belonging items; five mathematics-anxiety items (2012 and 2022, the two cycles with mathematics as the main domain); and five mathematics disciplinary-climate items (2012 and 2022). We analyse raw item responses, not cycle-specific scaled indices, and we test their comparability directly (Section 4).

Administration mode. The other nine systems moved from paper-based to computer-based assessment in 2015 or 2018. Viet Nam remained paper-based in all four cycles (Appendix Table A1); this removes an administration-mode transition but does not establish score comparability. OECD explicitly warns that its 2022 performance results are not comparable with previous cycles (OECD, 2023b). We therefore analyse questionnaire responses over time and achievement associations within cycles; no Vietnamese mathematics trajectory is drawn. Two data constraints are specific to Viet Nam. First, its 2018 performance data are not included in the public file, so 2018 contributes questionnaire indicators only. Second, the 2022 digital-habit items, which were part of the computer-based questionnaire only, are analysed for the eleven computer-based systems.

4. Method

Design-based estimation. Student-level descriptive and regression estimates use final student weights. Sampling variance uses 80 balanced repeated replication weights (Fay factor 0.5), and plausible-value variance (five in 2012, ten afterwards) is combined with Rubin's rules (OECD, 2009; Rubin, 1987). Continuous predictors are standardised within system and cycle. Planned questionnaire rotation creates missingness by design; this does not make all item nonresponse missing completely at random. Regressions use complete cases on their variables, descriptive estimates use available responses, DFL uses missing-covariate categories, and LightGBM handles missing predictors natively. Complete-case associations require appropriate assumptions about response selection; analytic sample sizes are reported. Across-system hypothesis tests use false-discovery-rate correction (Benjamini & Hochberg, 1995). CFA fit and machine-learning diagnostics are weighted descriptive estimates rather than BRR-adjusted hypothesis tests.

Measurement invariance. For Viet Nam we fitted one-factor multi-group confirmatory factor models to weighted item moments by maximum likelihood, separately for belonging (four cycles; residual covariances among the three negatively worded items to absorb the wording method effect) and mathematics anxiety (2012 vs 2022). We tested configural, metric and scalar invariance, applying the $\Delta\text{CFI} \geq -0.010$ criterion (Chen, 2007; Putnick & Bornstein, 2016), and estimated latent mean differences in units of the 2012 standard deviation.

Testing the individual-exposure account (H2). Within each system and cycle, we regressed loneliness (linear probability, percentage points), the belonging scale and mathematics anxiety on home computer, internet and smartphones, controlling for ESCS, gender, grade, grade repetition and books at home. We then decomposed the 2012–2022 change in Viet Nam with DiNardo–Fortin–Lemieux reweighting (DiNardo et al., 1996). The 2022 sample was reweighted to reproduce the 2012 distribution of (i) composition (gender, grade, repetition, books) and (ii) composition plus home computer and internet. The differences between the actual and counterfactual 2022 means give the parts of the change explained by composition and by access.

Digital distraction, 2022 (H3–H4). DIGDIST is the mean of two items on distraction by one's own and by peers' digital devices in mathematics lessons. The official 2022 DISCLIM index contains these items, so we built a non-digital disciplinary-climate index from the five classic items. Models M1/M2 predict mathematics (background controls; + non-digital climate), and models A1/A2 predict mathematics anxiety (background controls; + climate, belonging, bullying, teacher support).

School policies (H5). Phone bans are not randomly assigned. For each system we estimated school-level propensity scores for having a ban from school-mean ESCS, location, size and type, and applied stabilised inverse-probability weights (Rosenbaum & Rubin, 1983), with propensity probabilities clipped to .05/.95, in pooled models with system fixed effects and equal system weights.

Interpretable machine learning. LightGBM models (Ke et al., 2017) were trained per system and outcome. For performance, one model was trained for each plausible value; student weights were used and missing values were handled natively. Accuracy was assessed with five-fold cross-validation *grouped by school*, against a weighted linear benchmark. Predictor importance is the weighted mean absolute SHAP value (Lundberg & Lee, 2017). Separate cycle models provide exploratory prediction summaries. Their SHAP percentages have model-specific denominators and background distributions; they are not harmonised importance estimates across cycles.

Additional checks and calibration. We re-estimated belonging with B3 as the positive anchor, freeing the B1/B4/B6 intercepts across cycles, including a specification omitting 2018. Partial-scalar CFI is compared with the metric model, not with the already rejected full-scalar model. DFL sensitivity includes untrimmed weights, propensity clipping at .01/.99 and .05/.95, and density-ratio winsorisation at the 1st/99th and 5th/95th percentiles. Each stage is refitted with all 80 replicate weights separately for 2012 and 2022, summing independent-cycle variances. Weighted logistic fitting uses mean-one sampling weights to stabilise regularisation. A conditional permutation diagnostic jointly shuffles computer/internet labels 199 times within gender by ESCS-quintile strata; it is a model check, not a causal placebo. Reading is analysed as a comparison outcome rather than assumed to be unrelated to technology. For Viet Nam's 2022 LightGBM models we report weighted out-of-fold observed-on-predicted slopes, intercepts and decile reliability. A separate classifier for mathematics below Level 2 (420.07 points) reports weighted Brier scores against each fold's training-prevalence baseline (OECD, 2023c). All ten mathematics plausible values are fitted separately; target-missing cases are excluded. No validation fold contains schools from its training folds.

Measurement and response processes. Latent mean comparison requires comparable measurement structure, not merely identical wording (Meredith, 1993; Vandenberg & Lance, 2000). Partial invariance can be useful when a defensible invariant subset anchors comparisons (Byrne et al., 1989); freeing problematic parameters does not by itself validate a model. International-survey fit cutoffs depend on group structure and model conditions (Rutkowski & Svetina, 2014; Svetina & Rutkowski, 2017). Ordinal-invariance and differential-item-functioning approaches offer useful extensions beyond our continuous-moment approximation (Millsap & Yun-Tein,

2004; Svetina & Rutkowski, 2014). We consequently distinguish directly observed response proportions from claims about an invariant psychological construct.

Overlap, selection and model interpretation. Propensity methods require balance and adequate support as well as assumptions about unmeasured selection (Austin, 2009; Austin & Stuart, 2015; Stuart, 2010). Severe distributional separation can make population reweighting unstable; restricting support or using overlap weights defines a different target population (Crump et al., 2009; Li et al., 2018). We do not relabel winsorised DFL ratios as an identified overlap-population effect. SHAP attributions depend on feature dependence and how the background distribution is specified (Aas et al., 2021; Janzing et al., 2020); within-model attribution therefore cannot be promoted to a standardised between-cycle inequality measure.

Digital use, pandemic context and Vietnamese schooling. Reviews distinguish access and duration from use patterns, individual vulnerability and social context (Dienlin & Johannes, 2020; Odgers & Jensen, 2020; Odgers et al., 2020). Pandemic-era syntheses document distress and isolation concerns but do not identify the source of our PISA cohort differences (Loades et al., 2020; Racine et al., 2021; Samji et al., 2022). A mixed-method study of Hanoi high-school students reports academic, social and family concerns surrounding anxiety (Hoa et al., 2024); its local sample and GAD-7 outcome are not interchangeable with national PISA mathematics anxiety. The SMART Schools observational study adds recent evidence on restrictive policies, phone use and well-being (Goodyear et al., 2025); policies on paper, enforcement and students' overall use are distinct exposures. These sources motivate alternatives to a uniquely digital ecological explanation, without assigning those alternatives causal effects in our data.

5. Results

Unless specified otherwise, descriptive tables use available-case final-weighted estimates, regression tables use model-specific complete cases and final weights, and parenthesised SEs use all 80 Fay BRR replicates (plus Rubin combination for achievement). Any stars denote $*p < .05$, $**p < .01$, $***p < .001$; q denotes the Benjamini–Hochberg adjusted value. These conventions do not apply to CFA fit statistics, SHAP importance or prediction diagnostics. Table-specific sample sizes are unweighted counts.

5.1. Are the measures comparable over time?

Table 2 reports the original invariance models. Mathematics anxiety met scalar invariance between 2012 and 2022 ($\Delta\text{CFI} = -0.009$; $\text{CFI} = 0.958$). Its RMSEA was elevated, as can occur with few degrees of freedom (Kenny et al., 2015). The corresponding latent mean increased by **0.33 SD**. Belonging met metric invariance but failed scalar invariance. The targeted

partial-scalar re-estimation freeing all three negative-item intercepts improves CFI from 0.749 to 0.916, but still falls 0.060 below the metric model (Table A5). The 2022 latent contrast attenuates from −0.250 to −0.035 SD, or −0.042 SD when 2018 is omitted (Table A6). Thus the inference of a substantial common latent decline is sensitive to measurement assumptions; these failed-invariance estimates are diagnostic, not validated latent trends. Item-level changes remain informative (Table 3): negatively worded items move toward exclusion, while "other students seem to like me" increases. Complete item means, SDs and category frequencies are provided in Tables A7–A8; response-style changes remain a plausible explanation.

Table 2: Measurement invariance, Viet Nam (multi-group CFA, ML on weighted moments)

Scale (Viet Nam)	Model	χ^2	df	CFI	ΔCFI	RMSEA
Belonging	configural	348.4	24	0.981		0.052
Belonging	metric	451.5	39	0.976	-0.005	0.046
Belonging	scalar	4,417.1	54	0.749	-0.227	0.127
Belonging	partial scalar (free intercepts: B6)	4,298.0	51	0.755	+0.007	0.128
Belonging	partial scalar (free intercepts: B5)	3,123.1	51	0.823	+0.074	0.109
Belonging	partial scalar (free intercepts: B5, B2)	2,702.1	48	0.847	+0.098	0.105
Belonging (excl. 2018)	configural	222.2	18	0.985		0.048
Belonging (excl. 2018)	metric	298.7	28	0.980	-0.005	0.044
Belonging (excl. 2018)	scalar	3,479.8	38	0.743	-0.237	0.135
Belonging (excl. 2018)	partial scalar (free intercepts: B6)	3,439.8	36	0.746	+0.003	0.138
Belonging (excl. 2018)	partial scalar (free intercepts: B5)	2,248.2	36	0.835	+0.092	0.111
Belonging (excl. 2018)	partial scalar (free intercepts: B5, B2)	1,918.8	34	0.859	+0.116	0.106
Maths anxiety	configural	631.1	10	0.967		0.117
Maths anxiety	metric	651.1	14	0.967	-0.001	0.100
Maths anxiety	scalar	825.1	18	0.958	-0.009	0.099

Final-weighted item moments; complete cases. Belonging models include residual covariances among B1/B4/B6. Here ΔCFI is relative to the previous nested model, with earlier partial-scalar rows showing improvement over full scalar only; those positive differences do not establish invariance. The targeted partial-scalar check in Table A5 compares directly with metric. These ML fit statistics are not survey-adjusted tests.

Table 3: Viet Nam: belonging items, percentage giving the belonging response

Item (% giving the belonging response)	2012	2015	2018	2022	Δ 2012–22
Not an outsider (reversed)	94.5	95.3	72.4	78.5	-15.9
Make friends easily	92.0	91.6	80.0	85.4	-6.6
Feel I belong	82.7	80.8	73.7	82.4	-0.3
Not awkward (reversed)	88.8	82.7	76.7	70.5	-18.2
Other students like me	40.8	42.6	36.9	65.1	+24.3
Not lonely (reversed)	93.9	92.5	87.3	85.6	-8.3

5.2. A decade of change (RQ1)

Table 4 and Figure 2 trace Viet Nam over the decade. Home internet access rose from 29.2% to 89.6%, and households with three or more smartphones rose from 29% (2015) to 69%, the fastest diffusion among the ten systems. The share of students who agreed that they feel lonely at school more than doubled (6.1% to 14.4%). Viet Nam began the decade as the

least lonely system in the sample and recorded the largest increase (Table 5). Students who felt very tense doing mathematics homework rose from 33.3% to 45.5%, and those reporting noise and disorder in most mathematics lessons from 10.6% to 17.6%. Over the same period, disciplinary climate *improved* in Korea, Malaysia, Chinese Taipei and Singapore (Figure 3). RQ1 documents changes in reported item responses and in the anxiety model; it does not establish a general deterioration in mental health.

Table 4: Viet Nam, PISA 2012–2022: harmonised indicators (weighted; BRR standard errors in parentheses)

Indicator	2012	2015	2018	2022
Students (N)	4,959	5,826	5,377	6,068
Mathematics mean (not a linked trend)	511.3 (4.8)	494.5 (4.5)	–	469.4 (3.9)
Socio-economic gradient (points per ESCS unit)	29.4 (2.6)	26.9 (3.2)	–	27.6 (2.2)
Gender gap, girls minus boys (points)	-9.9	3.0	–	-10.3
Computer for school work at home (%)	39.6 (1.6)	44.2 (1.5)	41.6 (1.8)	53.4 (1.7)
Internet at home (%)	29.2 (1.7)	46.3 (1.5)	76.9 (1.4)	89.6 (0.8)
Three or more smartphones at home (%)	–	29.1 (1.3)	48.8 (1.6)	69.4 (1.2)
'I feel like I belong at school' (%)	82.7 (0.8)	80.8 (0.7)	73.7 (0.9)	82.4 (0.6)
'I feel lonely at school' (%)	6.1 (0.5)	7.5 (0.6)	12.7 (0.7)	14.4 (0.5)
Sense of belonging, six items (1–4)	3.02 (0.01)	3.07 (0.01)	2.81 (0.01)	2.91 (0.01)
Mathematics anxiety, five items (1–4)	2.54 (0.01)	–	–	2.67 (0.01)
'I get very tense doing maths homework' (%)	33.3 (1.2)	–	–	45.5 (0.9)
Disciplinary climate in maths (1–4, higher = better)	3.25 (0.01)	–	–	3.22 (0.01)
Noise and disorder in most/every maths lesson (%)	10.6 (0.8)	–	–	17.6 (0.7)

2018 performance data for Viet Nam are not in the public file. Mathematics means are separate cycle descriptions; Viet Nam 2022 is not comparable with earlier cycles.

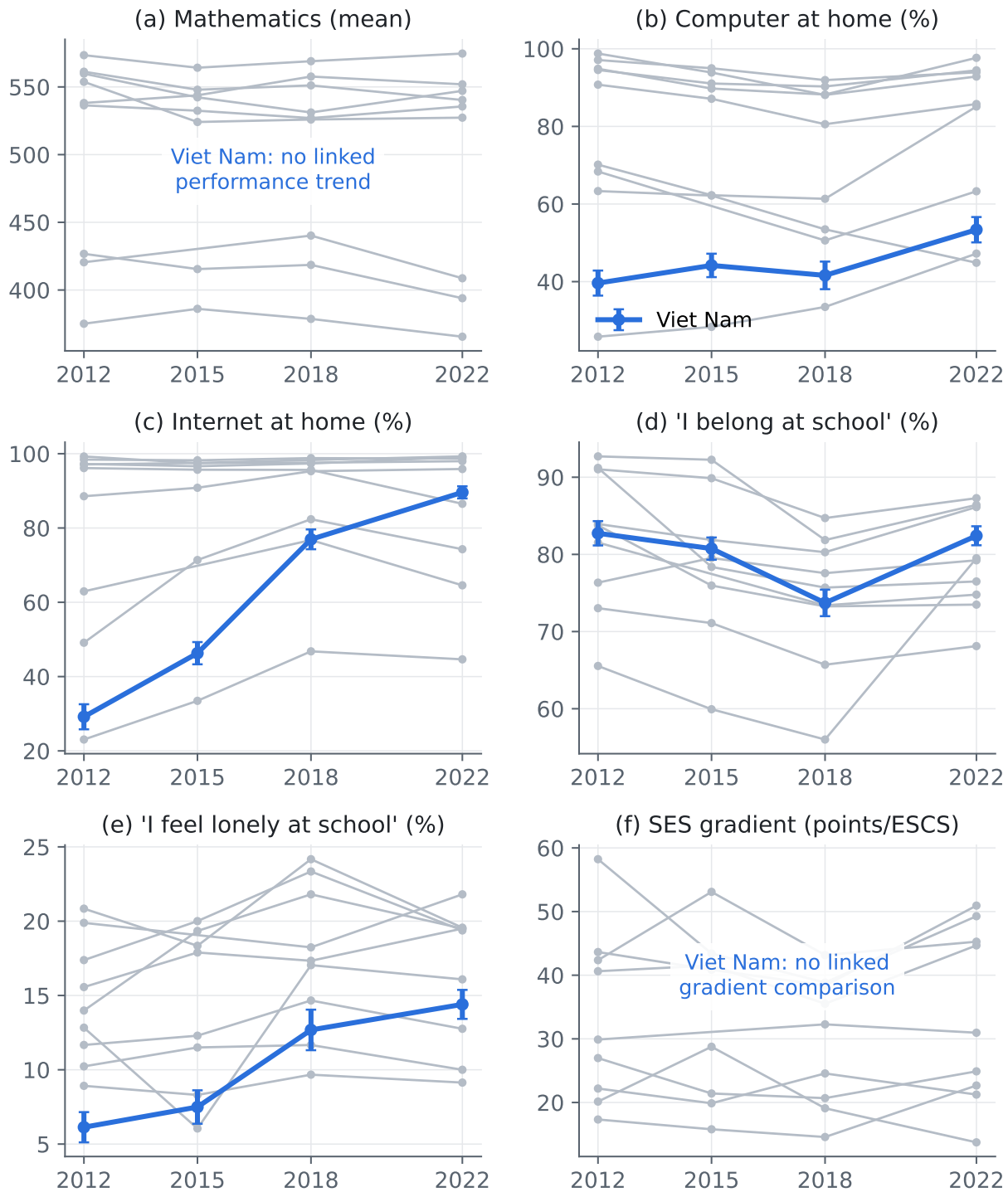


Figure 2: Trends 2012–2022: (a) mathematics in nine comparison systems only (Viet Nam is omitted because its 2022 score is not reliably comparable with earlier cycles), (b) computer access, (c) internet access, (d) belonging agreement, (e) loneliness, (f) SES gradient in the nine comparison systems only. No linked Vietnamese gradient comparison is inferred. Viet Nam is blue in questionnaire panels with 95% BRR intervals; other systems are grey. Pointwise intervals for cycle means are not confidence intervals for cross-cycle differences; link-adjusted mathematics contrasts are in Table A16. Total $n = 250,587$; cycle-specific samples are in Table 1. Available responses and final student weights are used.

Table 5: Change 2012–2022 in harmonised indicators (percentage points unless stated)

	Internet (pp)	Lonely (pp)	Belong (pp)	Maths anxiety (1–4)	Tense (pp)	Noise (pp)
Viet Nam	+60.4***	+8.3***	-0.3	+0.14***	+12.2***	+7.0***
Singapore	+2.2***	+3.9***	-10.3***	-0.04**	+0.1	-5.5***
Japan	+7.3***	-0.2	+2.2*	-0.01	-1.3	-4.9***
Korea	+1.9***	+0.2	+2.9*	-0.22***	-1.0	-20.5***
Chinese Taipei	-9.6***	+1.1	-3.7***	-0.02	+7.0***	-8.5***
Hong Kong (China)	-1.3***	+5.5***	-4.9***	+0.09***	+14.7***	-4.0***
Macao (China)	+0.3	+2.0*	+14.0***	+0.06***	+2.0*	+2.1*
Thailand	+25.2***	-1.3	-14.7***	-0.03	+2.7	-1.4
Malaysia	+1.6	+1.9	-6.7***	+0.08***	+7.2***	-12.4***
Indonesia	+21.6***	+3.3***	-6.3***	+0.16***	+16.9***	+6.1***

* $p < .05$, ** $p < .01$, *** $p < .001$ (independent samples, BRR standard errors).

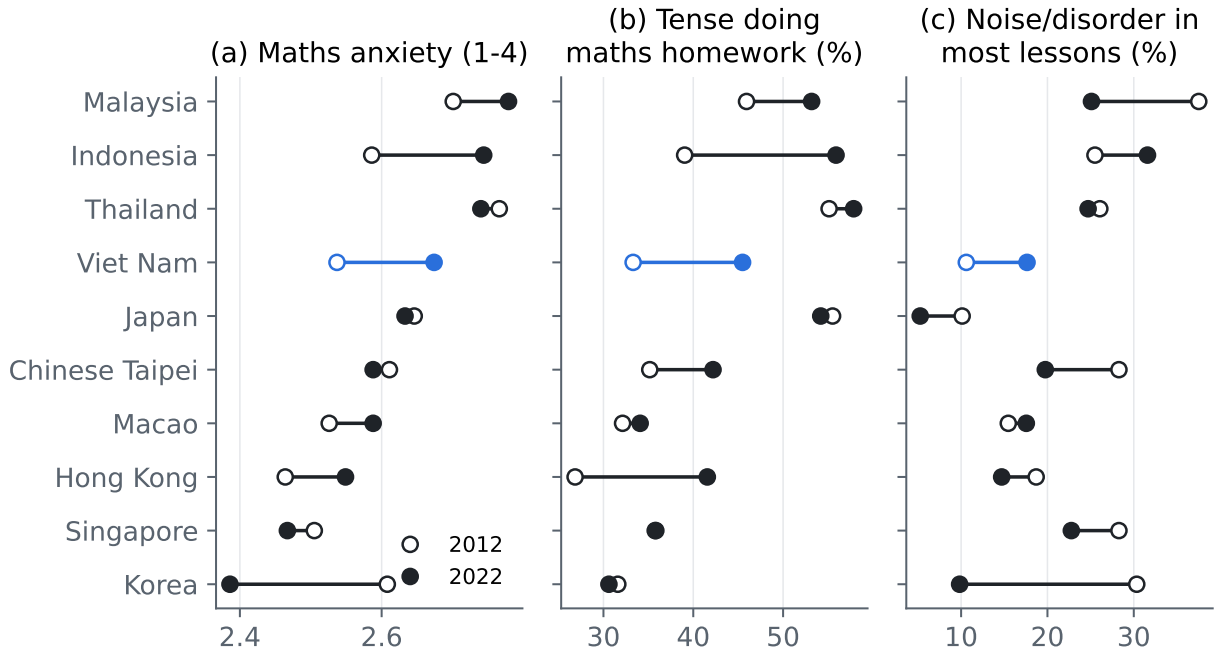


Figure 3: 2012 (open) versus 2022 (filled): (a) anxiety, (b) homework tension, (c) classroom noise. Ten systems; available responses and final student weights. Cycle samples are in Table 1; these panels show means without inferential intervals.

5.3. Is the trend explained by students' own digital access? (H2)

The measured home-access indicators provide little evidence for H2 in the adjusted within-cycle models (Table 6). Across the region, 91 system-cycle-indicator tests in the region, 25 significant at 5%, all negative : where access mattered, it was associated with *less* loneliness. In the updated DFL analysis, composition contributes −0.61 percentage points and access −1.24 points to the observed +8.27-point loneliness increase (Table 7). The access point estimate is

non-positive across all five clipping specifications (Table A9); its untrimmed approximate 95% interval is [−4.7, 2.2] percentage points, so the estimate is not evidence of equivalence. The access-stage effective sample size is only 367 of 6,068 students with untrimmed weights, rising to 637 with .05/.95 probability clipping and 1,123 with 5th/95th-percentile ratio winsorisation (Table A10). Limited overlap therefore warrants caution despite stable point-estimate direction. These are overlap-limited sensitivity results, and imprecise access components cannot demonstrate absence of an association; nonsignificance does not establish equivalence or exclude effects of usage intensity, content or unmeasured context. Conditional permutation diagnostics yield no unusual home-access coefficients under the selected null (Table A12). Reading and mathematics are both positively associated with ICT resources in 2022 (17.76 and 18.17 points per SD; Table A13); reading is therefore unsuitable as a technology-unrelated negative control.

The 2022 within-cycle access–achievement association raises an equity question alongside the uncertain DFL sensitivity results. It does not establish that inequality increased over time. H5 concerns school phone policies and self-regulation.

Table 6: Viet Nam: within-cycle associations of home digital access with well-being (SE in parentheses)

Outcome	Home access	2012	2015	2018	2022
Lonely (pp)	Computer	-1.4 (1.5)	0.7 (0.8)	-0.2 (1.2)	0.6 (1.4)
Lonely (pp)	Internet	0.6 (1.6)	0.0 (0.9)	0.1 (1.6)	-1.8 (1.9)
Lonely (pp)	Smartphone 3+	–	1.4 (0.8)	-0.5 (1.2)	-1.7 (1.2)
Belonging (1–4)	Computer	0.014 (0.024)	-0.020 (0.016)	0.004 (0.014)	0.029 (0.016)
Belonging (1–4)	Internet	0.029 (0.027)	0.020 (0.018)	0.017 (0.023)	0.050 (0.026)
Belonging (1–4)	Smartphone 3+	–	-0.006 (0.015)	0.020 (0.015)	0.025 (0.015)
Maths anxiety (1–4)	Computer	-0.018 (0.027)	–	–	0.005 (0.028)
Maths anxiety (1–4)	Internet	-0.040 (0.029)	–	–	0.017 (0.034)
Maths anxiety (1–4)	Smartphone 3+	–	–	–	0.015 (0.021)

Controls: ESCS, gender, grade, grade repetition, books at home. Lonely: percentage points; belonging and anxiety: scale points (1–4).

Table 7: Viet Nam: DiNardo–Fortin–Lemieux decomposition of the 2012–2022 change

Outcome	Change	Composition	Access	Residual
Loneliness (pp)	8.27 (0.72)	-0.61 (0.42)	-1.24 (1.77)	10.12 (2.00)
Belonging (1–4)	-0.11 (0.01)	-0.00 (0.01)	0.05 (0.03)	-0.16 (0.03)
Anxiety (1–4)	0.14 (0.02)	-0.00 (0.01)	0.01 (0.03)	0.13 (0.03)
Tension (pp)	12.18 (1.55)	-0.52 (0.51)	-0.50 (2.15)	13.20 (2.53)
Noise (pp)	7.03 (1.09)	-0.56 (0.45)	1.52 (1.99)	6.07 (2.29)

SE in parentheses; final student weights with untrimmed DFL ratios. Composition: gender, grade, repetition, books; access: computer/internet. Available outcomes; missing covariates form categories. All 80 Fay BRR replicates are refitted independently in each cycle. Residual = change not accounted for by measured composition/access, not an identified causal mechanism. Full sample and weight diagnostics accompany the code.

Access and achievement within the 2022 cohort. In Viet Nam in 2022, home-computer access was associated with 23.4 higher mathematics points after background adjustment. This association may reflect resources, learning opportunities and unmeasured selection; it is not the effect of providing a computer. Separately fitted cross-cycle SHAP summaries are retained only as exploratory model documentation in Figure A3. No change in an inequality parameter is estimated from them.

5.4. Digital distraction in 2022 (H3, H4)

In 2022, 14.3% of Vietnamese students reported digital distraction in most or every mathematics lesson, compared with 5.2% in Japan and 40.9% in the Philippines (Table 8, Figure 4). With background controls, distraction predicted lower mathematics scores in 12 of 13 systems (Viet Nam −4.1 points per SD). After non-digital disorder was controlled, the association disappeared in Viet Nam (+0.7) and was significant in only 6 of 13 systems after false-discovery-rate correction (Table 9, Figure 5). H3 is supported. Distraction predicted mathematics anxiety with background controls in 9 of 10 systems. After controlling for climate, belonging, bullying and teacher support, **Viet Nam was the only system** in which the association remained significant after correction (+0.08 SD, $q < .001$; Figure 6). The SHAP dependence curve shows that it is concentrated among students who are distracted in every lesson (Figure 7). H4 is supported for Viet Nam. In the eleven computer-based systems, students who turned off notifications in class scored 9–20 points higher per SD in 10 of 11 systems (Figure 8).

Table 8: Digital distraction, digital habits, mathematics and anxiety, PISA 2022

System	*N*	Distracted, most lessons (%) own devices	peers' devices	Online pressure (%)	Nervous w/o device (%)	Maths	ANXMAT
Viet Nam	6,068	14.2 (0.7)	12.7	–	–	469.4 (3.9)	0.45
Singapore	6,606	27.3 (1.1)	20.6	11.6	16.3	574.7 (1.2)	0.15
Japan	5,760	5.2 (0.5)	3.9	4.0	17.0	535.6 (2.9)	0.33
Korea	6,454	9.4 (0.6)	8.8	24.5	10.0	527.3 (3.9)	-0.05
Chinese Taipei	5,857	15.9 (1.0)	12.5	21.9	14.1	547.1 (3.8)	0.29
Hong Kong (China)	5,907	16.4 (0.9)	15.2	19.3	20.5	540.4 (3.0)	0.23
Macao (China)	4,384	13.3 (0.7)	11.6	23.1	23.6	551.9 (1.1)	0.28
Thailand	8,495	26.4 (0.9)	22.9	10.9	21.7	393.9 (2.7)	–
Malaysia	7,069	20.3 (1.0)	19.0	16.3	27.1	408.7 (2.4)	0.50
Indonesia	13,439	25.1 (0.9)	26.6	15.8	35.1	365.5 (2.4)	–
Philippines	7,193	40.9 (0.9)	37.5	29.7	23.9	354.7 (2.6)	–
Brunei Darussalam	5,576	11.5 (0.4)	10.5	23.6	25.6	442.1 (0.9)	0.67
Cambodia	5,279	19.2 (0.8)	20.8	–	–	336.4 (2.7)	0.50

"–": not administered (paper-based systems or national option).

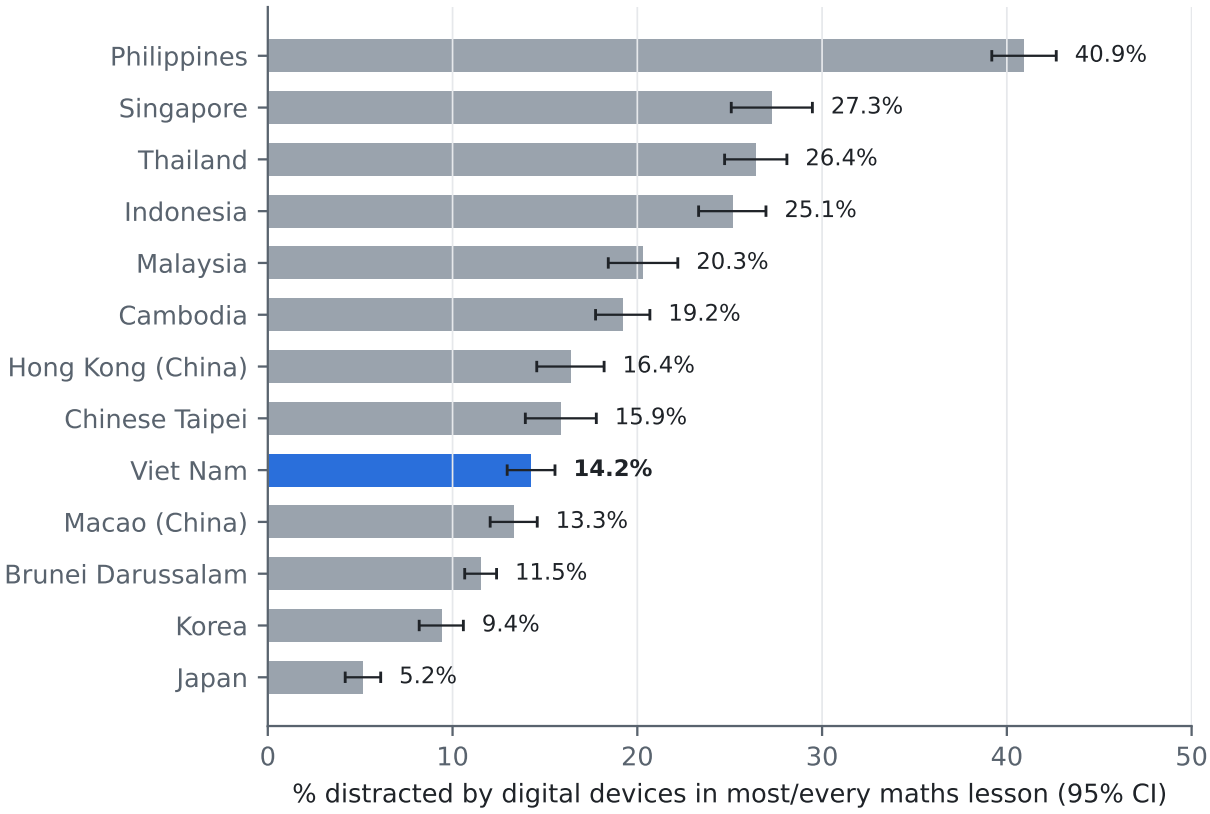


Figure 4: Students distracted by digital devices in most or every mathematics lesson: final-weighted percentages with 95% BRR intervals. Thirteen systems, base $n = 88,087$; percentages exclude missing item responses.

Table 9: Digital distraction (DIGDIST) coefficients, PISA 2022 (SE in parentheses)

	Mathematics (points per SD)		Maths anxiety (index per SD)	
System	M1	M2 (+climate)	A1	A2 (+climate, relations)
Viet Nam	-4.1* (1.9)	0.7 (2.3)	0.20*** (0.02)	0.08*** (0.02)
Singapore	-9.9*** (2.1)	0.9 (2.3)	0.11*** (0.02)	0.01 (0.02)
Japan	-7.4** (2.5)	1.1 (2.3)	0.09** (0.03)	0.00 (0.03)
Korea	-6.0** (2.1)	-7.6** (2.4)	0.08** (0.03)	0.03 (0.03)
Chinese Taipei	0.8 (3.3)	5.1 (3.9)	0.09** (0.03)	0.03 (0.03)
Hong Kong (China)	-10.7*** (2.8)	-3.8 (3.2)	0.13*** (0.03)	0.06* (0.03)
Macao (China)	-13.1*** (1.8)	-11.0*** (2.0)	0.06* (0.02)	-0.02 (0.03)
Thailand	-6.9*** (1.6)	2.3 (2.2)	–	–
Malaysia	-16.8*** (1.5)	-8.9*** (1.7)	0.05* (0.02)	-0.04 (0.03)
Indonesia	-6.5*** (1.4)	-3.1 (1.8)	–	–
Philippines	-5.8*** (1.2)	3.6** (1.2)	–	–
Brunei Darussalam	-13.6*** (1.8)	-7.3*** (1.8)	0.03 (0.02)	-0.00 (0.02)
Cambodia	-8.8*** (1.4)	-7.4*** (1.7)	0.15*** (0.01)	0.04* (0.02)

Controls: ESCS, gender, grade, grade repetition. * $p < .05$, ** $p < .01$, *** $p < .001$ (unadjusted). Benjamini–Hochberg significant at $q < .05$: M1: 12/13; M2: 6/13; A1: 9/10; A2: 1/10 .

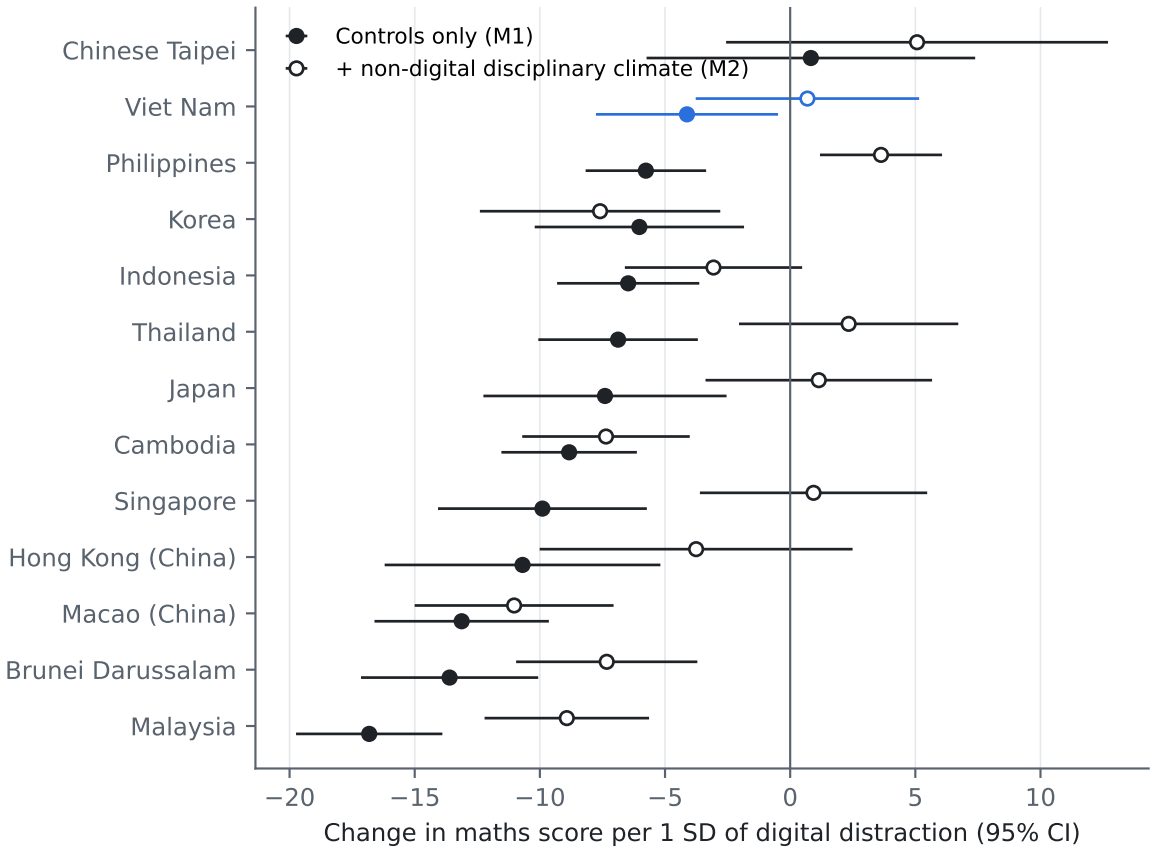


Figure 5: Digital distraction and mathematics before (M1) and after (M2) conditioning on non-digital disorder. Dots: coefficients; bars: 95% BRR/Rubin intervals using ten plausible values, final weights and complete cases. Thirteen systems; system sample sizes are in Table A15.

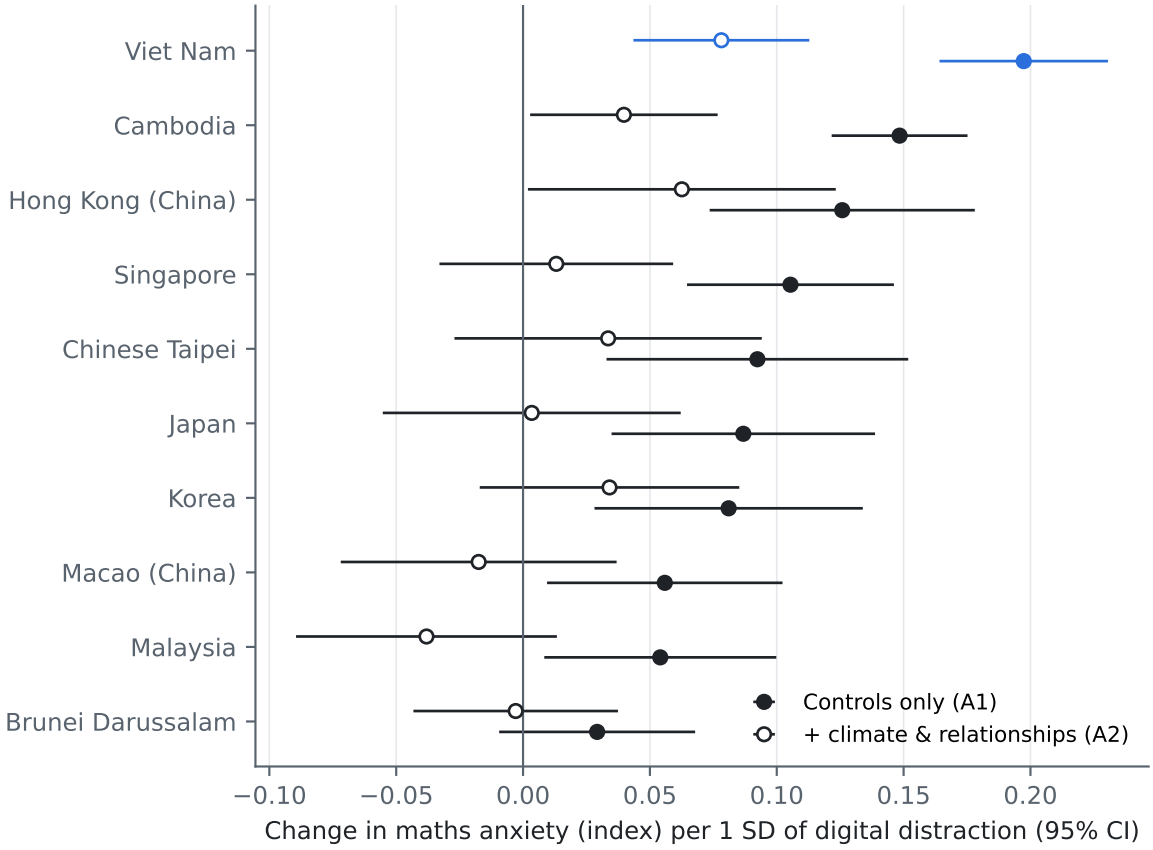


Figure 6: Digital distraction and mathematics anxiety before (A1) and after (A2) controls. Dots: coefficients; bars: 95% BRR intervals using final weights and complete cases. Ten systems; model sample sizes are in Table A15.

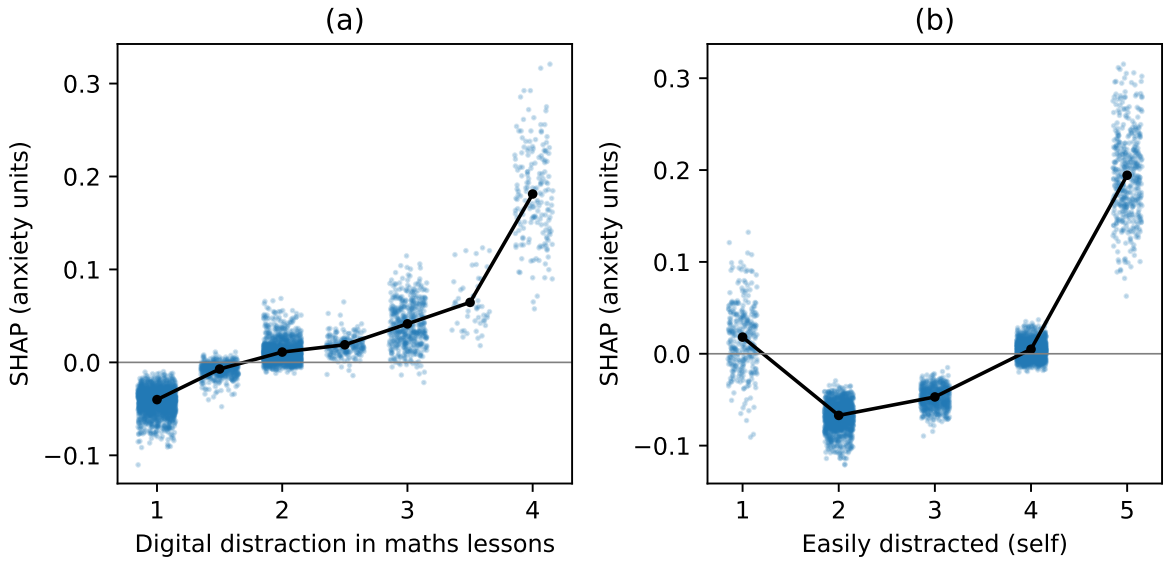


Figure 7: Viet Nam: SHAP dependence of mathematics anxiety on (a) digital distraction in mathematics lessons (1 = never, 4 = every lesson) and (b) "I get easily distracted" (1 = strongly disagree, 5 = strongly agree). Black line: unweighted mean SHAP value per response level; $n = 5,934$ Vietnamese students with observed anxiety. The fitted LightGBM model uses final student weights; this is a predictive dependence plot, not a causal contrast.

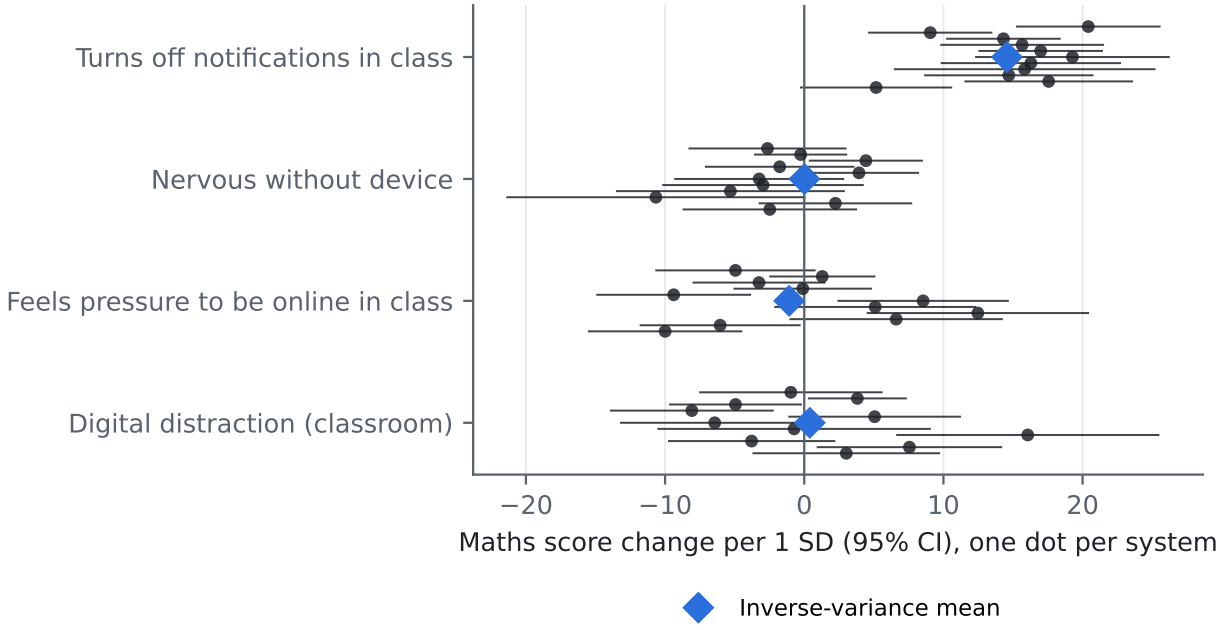


Figure 8: Digital habits and mathematics in eleven computer-based systems. Dots and bars: coefficients and 95% BRR/Rubin intervals from all ten plausible values; blue diamonds: inverse-variance means. Final weights and model-specific complete cases; total analytic $n = 13,339$, with system counts in Table A15 (M4).

5.5. School phone policies (H5)

Only 11.6% of Vietnamese students attended schools whose principal reported that phones were not allowed on the premises, compared with 64–81% in Hong Kong, Malaysia and Brunei (Table 10). In the naive pooled model, bans were associated with less distraction and with lower mathematics scores (Table 11). After inverse-probability weighting (Table 12), bans remained associated with less distraction (-0.08 SD, $p < .001$), whereas the achievement difference shrank to -1.9 points and was not significant. The attenuation is consistent with measured selection into bans; residual confounding and uncertainty prevent a causal interpretation. Bans were not associated with anxiety. Responsible-internet programmes were associated with lower anxiety in the pooled model, and in Viet Nam with less distraction. The phone-ban part of H5 is consistent with the distraction association; the self-regulation part is consistent with the achievement associations in the computer-based systems. Neither is a causal policy estimate.

Table 10: Students in schools with each device policy, PISA 2022 (%)

System	Phone ban (% students)	Co-created rules (%)	Responsible-internet programme (%)	SEL policy (%)
Viet Nam	11.7	78.7	70.8	66.1
Singapore	15.5	88.4	99.4	98.7
Japan	38.1	55.9	57.2	–
Korea	23.3	83.5	77.7	73.5
Chinese Taipei	45.8	77.5	68.1	44.2
Hong Kong (China)	67.0	59.8	86.3	60.9
Macao (China)	46.3	50.8	83.2	70.7
Thailand	12.4	90.1	57.9	85.0
Malaysia	64.4	60.2	61.2	70.0
Indonesia	43.5	85.4	76.7	79.0
Philippines	30.0	94.3	77.8	86.1
Brunei Darussalam	80.9	91.0	59.4	40.9
Cambodia	33.4	73.1	66.3	62.0

Table 11: School policies and student outcomes, regression adjustment (SE in parentheses)

	Pooled, 13 systems (system FE)			Viet Nam		
School policy	Distraction	Anxiety	Maths	Distraction	Anxiety	Maths
Phones not allowed on premises	-0.11*** (0.01)	0.00 (0.01)	-4.9* (1.9)	-0.04 (0.05)	-0.08 (0.05)	-9.2 (6.9)
Rules co-created with students	-0.03 (0.02)	-0.01 (0.02)	-0.7 (2.1)	-0.01 (0.04)	-0.11* (0.05)	-2.5 (6.9)
Responsible-internet programme	-0.03 (0.02)	-0.05* (0.02)	2.4 (2.3)	-0.11* (0.05)	0.01 (0.03)	-0.1 (7.5)
SEL policy	0.00 (0.02)	0.01 (0.02)	0.3 (2.3)	-0.03 (0.05)	-0.02 (0.04)	-1.0 (6.6)

Controls: student ESCS, school-mean ESCS, gender, grade, urban location; anxiety and mathematics models also control for DIGDIST. Singapore programme estimates rest on very few schools without the programme.

Table 12: School phone bans: pooled estimates with inverse-probability weighting (13 systems, system fixed effects)

Outcome	Unadjusted	IPW + student controls	N
Digital distraction (SD)	-0.082*** (0.011)	-0.084*** (0.011)	44,589
Maths anxiety (SD)	-0.017 (0.014)	-0.020 (0.013)	53,501
Mathematics (points)	-4.7 (3.0)	-1.9 (2.4)	85,299

Propensity score from school-mean ESCS, urban location, log school size and private management, estimated within each system; propensity probabilities clipped to .05/.95 before forming the weights.

6. Discussion

Well-being trends and measurement (RQ1). Viet Nam's rapid connectivity coincided with greater reported loneliness and a measurement-invariant increase in mathematics anxiety. The targeted partial-scalar analysis substantially attenuates the belonging contrast and still fails the invariance criterion. RQ1 is answered through item-response descriptions and anxiety-model comparisons, without a robust common latent belonging decline. Wording and response-style changes warrant caution in interpreting social exclusion.

Home access and equity (H2). The within-cycle and DFL results provide little support for the measured home-access account of loneliness. Five clipping choices document sensitivity rather than solve identification. Stage-2 ESS falls from 6,068 to 367 without clipping; 17.5% of estimated 2022 probabilities lie outside .05/.95. Low effective information can mask contributions, and clipping trades variance for bias without recovering unsupported population contrasts. Intensity, content, reverse causation and unobserved confounding remain unresolved. Aggregate trends (Twenge, 2026; Twenge et al., 2021) and small individual associations (Orben & Przybylski, 2019; Vuorre et al., 2021) can coexist. An ecological mechanism remains an untested possibility alongside examination pressure, family change and willingness to disclose distress. The 2022 home-computer achievement association motivates an equity question; its size must not be interpreted as growth in an inequality effect across non-comparable score scales.

Classroom climate and emotions (H3/H4). The distraction–achievement association attenuates after non-digital disorder is included, consistent with classroom attention and cognitive-load accounts (Sweller, 1988). Conditioning on a concurrent climate measure does not identify mediation, because temporal ordering and confounding of mediator and outcome are unresolved. The remaining distraction–anxiety association in Viet Nam is consistent with control-value theory (Pekrun, 2006). Cross-system significance differences do not themselves prove different effect sizes; interactions would be required for that claim. The regression reliability plots show reasonable mathematics calibration and an anxiety calibration slope below

one. SHAP describes prediction under correlated features, not mechanisms or causal variable importance.

Phone policies and self-regulation (H5). Bans are associated with less distraction after adjustment, while adjusted achievement and anxiety differences remain statistically uncertain. Propensity weighting addresses measured school characteristics only. Turning off notifications is positively associated with achievement in most computer-based systems, but behavioural selection may explain part of this pattern. Classroom management and self-regulation merit prospective evaluation alongside Viet Nam's conditional phone rule (Ministry of Education and Training of Viet Nam, 2020).

Pandemic and concurrent policy changes. PISA has no observations for 2020/2021 in this dataset. The loneliness increase was already 6.55 percentage points in 2012–2018, followed by 1.72 points in 2018–2022; belonging agreement decreased in the first interval but recovered in the second (Table A11). Thus COVID-19 alone cannot explain the earlier item trend, but the later interval does not isolate a closure effect. OECD's PISA 2022 analysis places pandemic disruptions alongside longer-term changes (OECD, 2023d). Viet Nam also adopted the 2018 general curriculum, with grade 10 scheduled from 2022/2023, and changed national graduation/university-entry assessment from 2015 (Government of Viet Nam, 2015; Ministry of Education and Training of Viet Nam, 2018). These reforms are concurrent context, not estimated causes; implementation dates and individual curriculum exposure must be checked before attributing the 2022 observations to the new grade-10 programme. Examination pressure and changing instruction remain alternative explanations.

Methodological contribution and recent evidence. The contribution combines multi-cycle questionnaire harmonisation with measurement-invariance tests, sequential DFL sensitivity, and within-cycle LightGBM/SHAP model documentation in ten East and Southeast Asian systems. This regional design should not be described as a global sample. Recent evidence includes OECD's PISA-2022 screen-time analysis (OECD, 2024), a 2024 adolescent device-use/achievement meta-analysis (Wang et al., 2024), and a 2025 systematic review of digital distractions (Martin et al., 2025). The latter already recognises technology, personal and environmental factors. Our addition is the joint temporal, measurement and distributional analysis of PISA data; ecological framing itself is not new. The 2024 meta-analysis is not exclusively a PISA-2022 synthesis, and the 2025 review is not a meta-analysis.

Implications and limitations. Monitor item-specific well-being alongside digital inclusion; evaluate classroom climate and self-regulation interventions; and prioritise equitable learning resources. These are research and policy priorities, not demonstrated treatment benefits.

Repeated cross-sections, complete-case selection, coarse access measures and concurrent outcomes limit causal inference. DFL accounts for measured distributional shifts only. Belonging remains non-invariant, and weighted-moment continuous-item CFA is an approximation to ordinal responses. The near-zero partial-scalar contrast neither proves stable true belonging nor proves that loneliness is a measurement artifact: both the full- and partial-scalar mean models fail. Changed negative-item response style is one plausible explanation, not an identified cause. Predictive calibration is internal school-grouped validation, without external or prospective validation. Viet Nam's 2022 performance scores are not reliably linked to its earlier cycles, and its 2018 performance values are unavailable; no Vietnamese achievement trend is inferred. For eligible comparison systems, the separate mathematics-difference sensitivity adds squared link error to sampling/PV variance (Table A16). Principal-reported restrictions do not measure compliance or enforcement; school management, disciplinary culture and teacher quality are unmeasured possible confounders. IPW targets adjustment for included covariates; balance is not guaranteed and unmeasured confounding remains. It is not a quasi-experiment. The small number of systems further limits inference.

7. Conclusion

For RQ1, Vietnamese students reported more loneliness and mathematics anxiety, but the partial-scalar check weakens a broad latent belonging-decline claim. For H2, overlap-limited reweighting yields uncertain access components and cannot establish that access is irrelevant; the 2022 within-cycle achievement association raises an equity question without establishing a temporal increase. For H3/H4, the adjusted patterns connect distraction with classroom climate and anxiety without identifying mediation. For H5, phone bans correlate with less distraction and self-regulation correlates with achievement, while causal policy effects remain unresolved. Longitudinal and intervention evidence is needed to guide a well-being and equity agenda for digital education.

Ethics statement

The study uses publicly released, de-identified secondary data. No new data were collected and no ethical approval was required.

Data and code availability

The international student and school files are available from the OECD PISA database, with earlier cycles linked in the accompanying code documentation. Context indicators come from World Bank WDI, WHO GHO and UNESCO PEER. A publication repository folder containing code, aggregate results, figure sources and LaTeX has been prepared locally for release.

A public code URL/DOI has not yet been assigned and must be added before submission; individual-level data are obtained from the original providers rather than redistributed in this folder.

CRediT authorship contribution statement

Author contributions are withheld for anonymous review.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI use

OpenAI Codex was used to assist with programming, robustness analyses, language editing and LaTeX preparation. It was not used to generate student observations or substitute for empirical estimates. The authors are responsible for reviewing the analyses, references and final text and for the content of the submitted article.

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Appendix A. Supplementary tables and figures

Table A1: PISA administration mode by system and cycle

System	2012	2015	2018	2022
Hong Kong (China)	paper	computer	computer	computer
Indonesia	paper	paper	computer	computer
Japan	paper	computer	computer	computer
Korea	paper	computer	computer	computer
Macao (China)	paper	computer	computer	computer
Malaysia	paper	–	computer	computer
Singapore	paper	computer	computer	computer
Chinese Taipei	paper	computer	computer	computer
Thailand	paper	computer	computer	computer
Viet Nam	paper	paper	paper	paper

Table A2: Earlier diagnostic latent-mean specifications, Viet Nam (2012 SD units; invariance failed)

Scale	Model	2015	2018	2022
Belonging	scalar	+0.18	-0.66	-0.25
Belonging	partial scalar (free intercepts: B6)	+0.15	-0.71	-0.26
Belonging	partial scalar (free intercepts: B5)	+0.21	-0.70	-0.39
Belonging	partial scalar (free intercepts: B5, B2)	+0.41	-0.92	-0.65
Belonging (excl. 2018)	scalar	+0.16	–	-0.25
Belonging (excl. 2018)	partial scalar (free intercepts: B6)	+0.14	–	-0.25
Belonging (excl. 2018)	partial scalar (free intercepts: B5)	+0.20	–	-0.38
Belonging (excl. 2018)	partial scalar (free intercepts: B5, B2)	+0.37	–	-0.66
Maths anxiety	scalar	–	–	+0.33

Earlier belonging specifications are shown for comparison and do not validate latent trends. The targeted negative-item analysis in Tables A5–A6 supersedes any claim of a substantial robust decline.

Table A3: Out-of-sample R^2 (school-grouped five-fold CV), PISA 2022

System	Mathematics		Maths anxiety	
	LightGBM	Linear	LightGBM	Linear
Viet Nam	0.371	0.301	0.199	0.197
Singapore	0.522	0.464	0.256	0.268
Japan	0.441	0.398	0.167	0.173
Korea	0.472	0.416	0.205	0.199
Chinese Taipei	0.524	0.453	0.213	0.211
Hong Kong (China)	0.393	0.315	0.297	0.303
Macao (China)	0.474	0.461	0.251	0.270
Thailand	0.393	0.269	–	–
Malaysia	0.556	0.471	0.179	0.183
Indonesia	0.364	0.248	–	–
Philippines	0.562	0.455	–	–
Brunei Darussalam	0.542	0.452	0.107	0.131
Cambodia	0.435	0.296	0.134	0.137

Table A4: System-level context, 2022 (World Bank, WHO, UNESCO PEER, PISA)

System	Internet users (%)	GDP pc (PPP, $k)	National phone policy (PEER)	Suicide rate	Distracted (%)	School ban (%)
Viet Nam	78.6	13.0	National conditional restriction	7.3	14.2	11.7
Singapore	96.0	134.8	School discretion	8.1	27.3	15.5
Japan	84.9	46.8	Not documented in PEER	17.4	5.2	38.1
Korea	97.2	53.2	School discretion	27.5	9.4	23.3
Chinese Taipei	–	–	Not covered by PEER	–	15.9	45.8
Hong Kong (China)	95.6	64.0	School discretion	–	16.4	67.0
Macao (China)	89.5	59.9	School discretion	–	13.3	46.3
Thailand	88.0	20.8	School discretion	16.6	26.4	12.4
Malaysia	97.4	32.1	No national rule found	5.7	20.3	64.4
Indonesia	66.5	13.3	No national rule	1.2	25.1	43.5
Philippines	75.2	9.5	No national rule found	3.5	40.9	30.0
Brunei Darussalam	99.0	76.4	National ban (bringing phones to school)	3.0	11.5	80.9
Cambodia	65.5	6.5	School discretion	4.5	19.2	33.4

Suicide rate: per 100,000, both sexes, all ages, 2021 (WHO). Across systems, distraction was not related to internet penetration ($\rho = -0.29$, $n = 12$) or to the share of schools banning phones ($\rho = -0.28$, $n = 13$). PEER coding with verbatim evidence: `data/raw/policy/peer_phone_policy_coding.csv`.

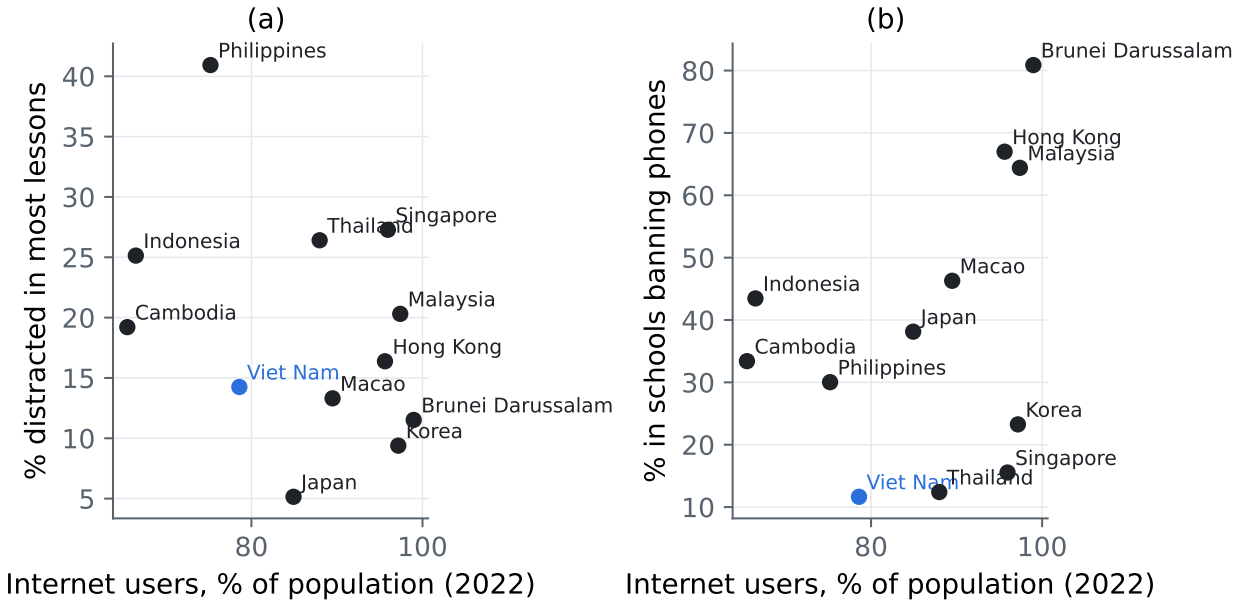


Figure A1: System context: (a) internet users versus distraction; (b) internet users versus phone bans. Available systems: $n = 12$ and $n = 13$. Student aggregates use final weights; system points are descriptive and unweighted.

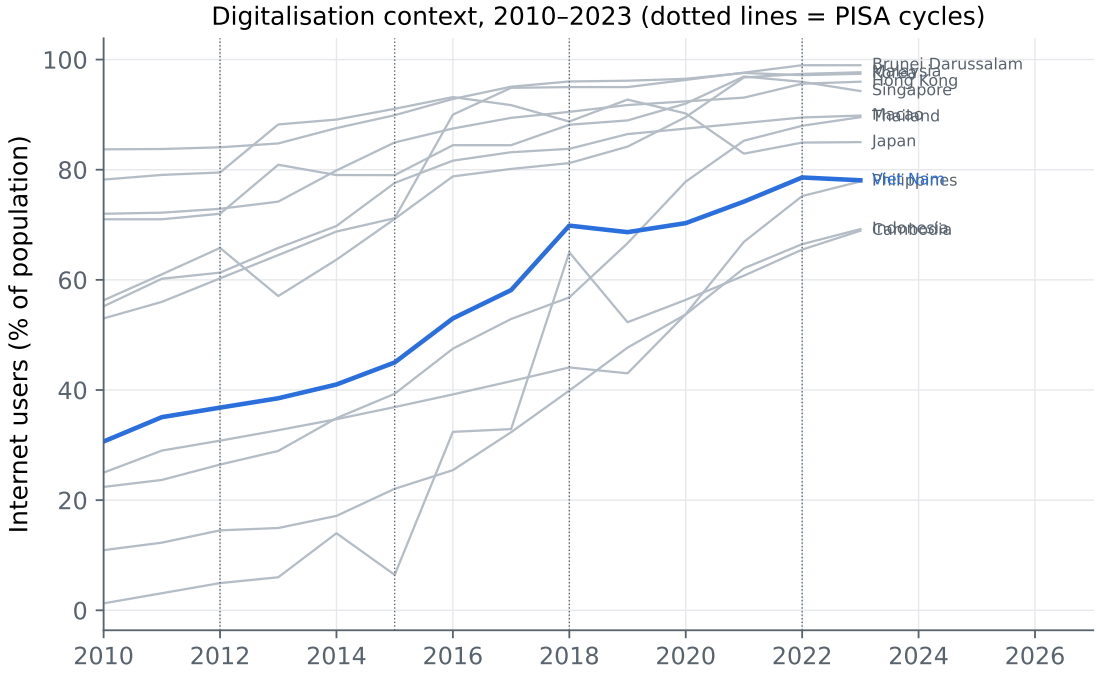


Figure A2: Internet users (% of population), 2010–2023 (World Bank).

Appendix A.1. Exploratory model documentation

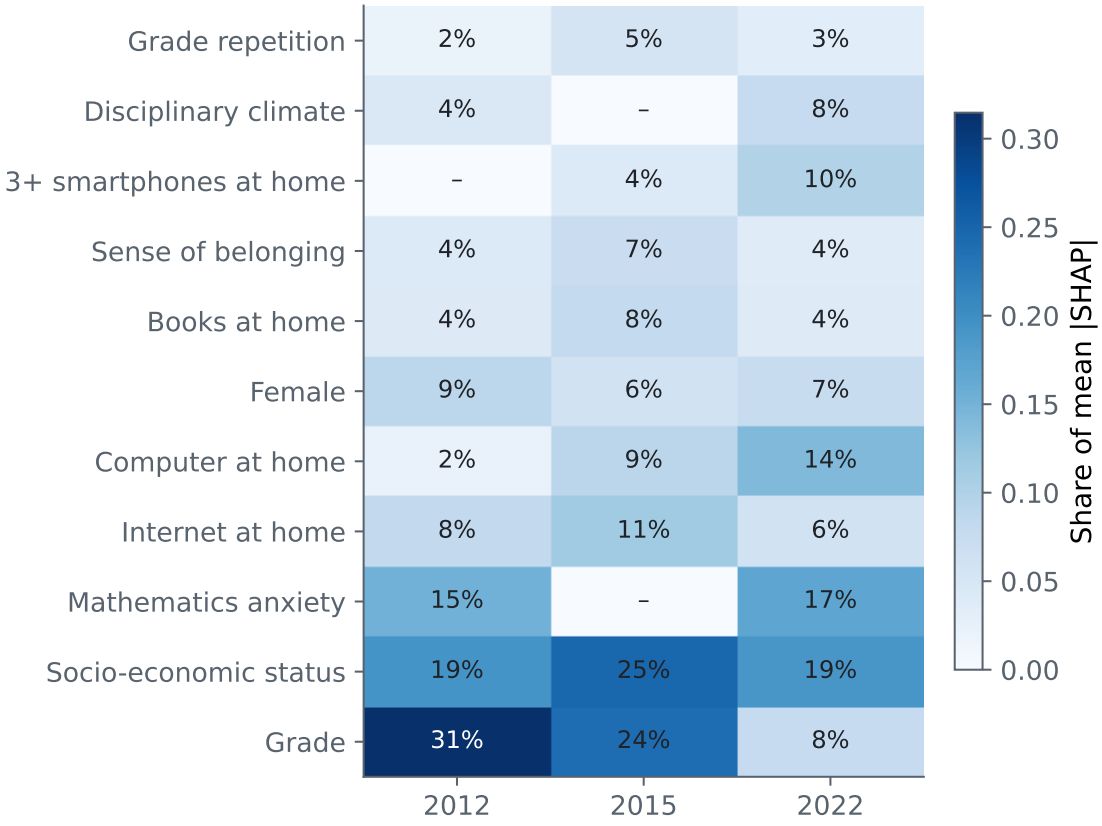


Figure A3: Viet Nam: separately normalised SHAP summaries for harmonised predictors of mathematics, by cycle (school-grouped CV $R^2 = 0.36$, 0.21, 0.23; $n = 4,959$, 5,826 and 6,068 in 2012, 2015 and 2022). Final-weighted LightGBM fits use native missing-predictor handling. Each model has its own feature covariance, availability and background distribution. These summaries are not a common cross-cycle importance scale and cannot demonstrate increasing digital inequality.

Appendix A.2. Additional measurement, sensitivity and calibration checks

Table A5: Belonging: targeted partial-scalar invariance, Viet Nam

Cycles	Model	n	df	CFI	RMSEA	ΔCFI
Four	Configural	20,213	24	0.981	0.052	–
Four	Metric	20,213	39	0.976	0.046	–
Four	Scalar	20,213	54	0.749	0.127	-0.227
Four	Partial (negative free)	20,213	45	0.916	0.080	-0.060
Three	Configural	14,920	18	0.985	0.048	–
Three	Metric	14,920	28	0.980	0.044	–
Three	Scalar	14,920	38	0.743	0.135	-0.237
Three	Partial (negative free)	14,920	32	0.905	0.089	-0.075

Final student weights; complete cases on all six items. Three-cycle models omit 2018. B3 (positive belonging) anchors the factor; B1, B4 and B6 intercepts are free across groups, with correlated residuals among these negative items. All models converged. Scalar and partial-scalar ΔCFI are relative to the metric model; neither meets the −0.010 criterion. Weighted-moment ML treats the four response categories as continuous; fit statistics are not survey-adjusted chi-square tests.

Table A6: Belonging: sensitivity of the 2022 latent mean relative to 2012

Cycles	Specification	2012 SD units
Four	Full scalar	-0.250
Four	Partial (negative free)	-0.035
Three	Full scalar	-0.246
Three	Partial (negative free)	-0.042

The full-scalar specification is a diagnostic comparison and fails invariance. The targeted partial-scalar models also fail; these point estimates do not establish a common latent-mean change. No design-based confidence interval is claimed.

Table A7: Belonging: item distributions by cycle, Viet Nam

Item	Year	n	Missing%	Mean	SD	1%	2%	3%	4%
B1	2012	3,301	33.4	3.26	0.60	1.6	4.0	61.7	32.7
B2	2012	3,301	33.4	3.13	0.55	0.5	7.5	69.9	22.0
B3	2012	3,281	33.8	2.98	0.61	1.2	16.1	66.2	16.5
B4	2012	3,294	33.6	3.12	0.62	1.2	10.1	63.9	24.8
B5	2012	3,299	33.5	2.36	0.68	8.2	50.9	37.5	3.3
B6	2012	3,295	33.6	3.28	0.60	1.0	5.2	59.0	34.9
B1	2015	5,796	0.5	3.51	0.62	0.9	3.7	39.1	56.3
B2	2015	5,803	0.4	3.17	0.60	1.4	6.9	65.1	26.6
B3	2015	5,772	0.9	2.93	0.66	3.1	16.1	65.4	15.4
B4	2015	5,798	0.5	3.05	0.69	2.1	15.2	58.6	24.1
B5	2015	5,793	0.6	2.36	0.73	10.5	46.9	38.3	4.3
B6	2015	5,798	0.5	3.40	0.69	2.2	5.3	42.5	50.0
B1	2018	5,341	0.7	2.82	0.74	4.9	22.7	57.3	15.0
B2	2018	5,342	0.7	2.92	0.64	2.5	17.5	65.5	14.5
B3	2018	5,324	1.0	2.80	0.64	3.1	23.2	64.4	9.4
B4	2018	5,337	0.7	2.87	0.67	3.3	20.0	62.9	13.8
B5	2018	5,333	0.8	2.32	0.68	8.8	54.4	33.5	3.4
B6	2018	5,340	0.7	3.13	0.70	3.3	9.4	58.7	28.7
B1	2022	5,986	1.4	2.95	0.80	6.7	14.8	55.8	22.8
B2	2022	6,011	0.9	3.05	0.72	4.7	10.0	61.3	24.1
B3	2022	5,987	1.3	2.95	0.66	3.5	14.1	66.7	15.8
B4	2022	6,002	1.1	2.77	0.72	5.3	24.2	58.8	11.8
B5	2022	5,982	1.4	2.67	0.69	5.4	29.5	57.6	7.5
B6	2022	6,003	1.1	3.08	0.79	6.7	7.7	56.6	29.0

Available-case means, population SDs and category percentages use final student weights; missing percentage is unweighted relative to the Vietnamese cycle sample. B1 outsider, B2 friends, B3 belonging, B4 awkward, B5 liked, B6 lonely; B1/B4/B6 are reversed, so higher B responses indicate more belonging. A1 worry about difficulty, A2 tense with homework, A3 nervous with problems, A4 helpless, A5 worry about poor grades; higher A responses indicate more anxiety. Items A1–A5 are available in 2012/2022 only. SD is dispersion, not a standard error.

Table A8: Mathematics anxiety: item distributions by cycle, Viet Nam

Item	Year	n	Missing%	Mean	SD	1%	2%	3%	4%
A1	2012	3,295	33.6	2.81	0.67	2.7	25.2	60.2	11.9
A2	2012	3,260	34.3	2.27	0.69	10.1	56.6	29.3	4.0
A3	2012	3,288	33.7	2.40	0.72	8.7	47.6	38.6	5.1
A4	2012	3,290	33.7	2.34	0.68	8.2	52.9	35.1	3.7
A5	2012	3,289	33.7	2.85	0.80	6.2	22.3	52.4	19.2
A1	2022	5,938	2.1	2.91	0.80	4.9	21.8	50.7	22.5
A2	2022	5,928	2.3	2.50	0.81	7.9	46.6	33.3	12.2
A3	2022	5,908	2.6	2.54	0.82	8.1	42.8	36.1	13.0
A4	2022	5,923	2.4	2.38	0.82	10.6	52.6	25.5	11.3
A5	2022	5,924	2.4	3.05	0.78	4.8	13.9	53.2	28.1

Available-case means, population SDs and category percentages use final student weights; missing percentage is unweighted relative to the Vietnamese cycle sample. B1 outsider, B2 friends, B3 belonging, B4 awkward, B5 liked, B6 lonely; B1/B4/B6 are reversed, so higher B responses indicate more belonging. A1 worry about difficulty, A2 tense with homework, A3 nervous with problems, A4 helpless, A5 worry about poor grades; higher A responses indicate more anxiety. Items A1–A5 are available in 2012/2022 only. SD is dispersion, not a standard error.

Table A9: DFL sensitivity to propensity and density-ratio clipping, Viet Nam

Outcome	Weights	Composition (SE)	Access (SE)	Residual (SE)
Loneliness (pp)	Untrimmed	-0.6 (0.4)	-1.2 (1.8)	10.1 (2.0)
Belonging (1–4)	Untrimmed	-0.00 (0.01)	0.05 (0.03)	-0.16 (0.03)
Anxiety (1–4)	Untrimmed	-0.00 (0.01)	0.01 (0.03)	0.13 (0.03)
Tension (pp)	Untrimmed	-0.5 (0.5)	-0.5 (2.1)	13.2 (2.5)
Noise (pp)	Untrimmed	-0.6 (0.5)	1.5 (2.0)	6.1 (2.3)
Loneliness (pp)	PS .01/.99	-0.6 (0.4)	-1.2 (1.8)	10.1 (2.0)
Belonging (1–4)	PS .01/.99	-0.00 (0.01)	0.05 (0.03)	-0.16 (0.03)
Anxiety (1–4)	PS .01/.99	-0.00 (0.01)	0.01 (0.03)	0.13 (0.03)
Tension (pp)	PS .01/.99	-0.5 (0.5)	-0.5 (2.1)	13.2 (2.5)
Noise (pp)	PS .01/.99	-0.6 (0.5)	1.5 (2.0)	6.1 (2.3)
Loneliness (pp)	PS .05/.95	-0.6 (0.4)	-1.0 (1.3)	9.8 (1.6)
Belonging (1–4)	PS .05/.95	-0.00 (0.01)	0.04 (0.02)	-0.15 (0.03)
Anxiety (1–4)	PS .05/.95	-0.00 (0.01)	0.00 (0.02)	0.14 (0.03)
Tension (pp)	PS .05/.95	-0.5 (0.5)	-1.6 (1.6)	14.3 (2.1)
Noise (pp)	PS .05/.95	-0.6 (0.5)	1.3 (1.4)	6.3 (1.8)
Loneliness (pp)	Ratio 1/99 pct	-0.5 (0.4)	-1.3 (1.6)	10.0 (1.9)
Belonging (1–4)	Ratio 1/99 pct	-0.00 (0.01)	0.04 (0.03)	-0.15 (0.03)
Anxiety (1–4)	Ratio 1/99 pct	-0.00 (0.01)	0.01 (0.03)	0.13 (0.03)
Tension (pp)	Ratio 1/99 pct	-0.5 (0.5)	-1.0 (2.0)	13.6 (2.4)
Noise (pp)	Ratio 1/99 pct	-0.4 (0.4)	1.4 (1.7)	6.0 (2.1)
Loneliness (pp)	Ratio 5/95 pct	-0.0 (0.2)	-1.6 (1.1)	9.8 (1.4)
Belonging (1–4)	Ratio 5/95 pct	-0.01 (0.00)	0.04 (0.01)	-0.15 (0.02)
Anxiety (1–4)	Ratio 5/95 pct	0.00 (0.00)	-0.00 (0.02)	0.14 (0.03)
Tension (pp)	Ratio 5/95 pct	0.0 (0.2)	-2.0 (1.3)	14.2 (2.1)
Noise (pp)	Ratio 5/95 pct	-0.1 (0.2)	0.6 (1.0)	6.5 (1.4)

Final student weights multiplied by sequential reweighting ratios. PS clips probabilities; ratio clipping winsorises the density ratio at empirical percentiles, without discarding observations. Composition: gender, grade, repetition, books; access adds computer/internet. Missing covariates form separate categories; outcomes use available cases. All 80 Fay BRR replicates are refitted separately in each independent cycle (160 fits per stage/specification); variances are summed across cycles. pp = percentage points. Residual is not a causal effect.

Table A10: DFL overlap and effective sample size diagnostics, 2022 cohort

Weights	Stage	n	ESS	Min PS	Max PS	Outside .05/.95%
Untrimmed	1	6,068	3789	0.130	0.726	0.0
Untrimmed	2	6,068	367	0.009	0.977	17.5
PS .01/.99	1	6,068	3789	0.130	0.726	0.0
PS .01/.99	2	6,068	368	0.009	0.977	17.5
PS .05/.95	1	6,068	3789	0.130	0.726	0.0
PS .05/.95	2	6,068	637	0.009	0.977	17.5
Ratio 1/99 pct	1	6,068	4013	0.130	0.726	0.0
Ratio 1/99 pct	2	6,068	426	0.009	0.977	17.5
Ratio 5/95 pct	1	6,068	4799	0.130	0.726	0.0
Ratio 5/95 pct	2	6,068	1123	0.009	0.977	17.5

Stage 1 uses composition, stage 2 adds access. ESS is $(\sum w)^2/\sum w^2$ after multiplying the final student weight by the reweighting ratio, across all 6,068 students before excluding outcome missingness. PS ranges and the outside percentage refer to original estimated probabilities; the latter is an unweighted percentage. Limited overlap can inflate uncertainty even when point estimates are stable.

Table A11: Trends before and across the pandemic interval, Viet Nam

Interval	Outcome	Change	SE	Per year
2012–2018	Loneliness (pp)	6.5	0.9	1.1
2012–2018	Belonging agreement (pp)	-9.0	1.2	-1.5
2012–2018	Belonging (1–4)	-0.21	0.01	-0.04
2018–2022	Loneliness (pp)	1.7	0.9	0.4
2018–2022	Belonging agreement (pp)	8.7	1.1	2.2
2018–2022	Belonging (1–4)	0.10	0.01	0.03

Final student weights and all 80 BRR replicates; independent-cycle variances are summed. PISA observations are not available in 2020/2021, so interval differences cannot identify a pandemic effect. Agreement proportions use available responses; raw scale means use the harmonised scale.

Table A12: Conditional permutation diagnostic for home access, Viet Nam

Year	Outcome	Access	n	Coefficient	Null mean	Null SD	p
2012	Loneliness (pp)	Computer	3,058	-1.417	0.079	1.329	0.285
2012	Loneliness (pp)	Internet	3,058	0.561	-0.396	1.419	0.690
2012	Anxiety (1–4)	Computer	3,060	-0.018	0.008	0.029	0.505
2012	Anxiety (1–4)	Internet	3,060	-0.040	0.004	0.032	0.195
2022	Loneliness (pp)	Computer	5,677	0.160	0.581	1.126	0.880
2022	Loneliness (pp)	Internet	5,677	-2.077	0.379	1.686	0.210
2022	Anxiety (1–4)	Computer	5,617	0.003	0.005	0.022	0.875
2022	Anxiety (1–4)	Internet	5,617	0.010	0.008	0.030	0.710

Joint computer/internet labels are shuffled 199 times within gender by ESCS-quintile strata, holding outcomes and survey weights fixed. WLS controls ESCS, gender, grade, repetition and books; complete cases are used. The two-sided empirical p uses an add-one correction. This diagnostic is conditional on the chosen strata and does not test absence of causal effects or replace design-based inference.

Table A13: Reading as a comparison outcome, Viet Nam, PISA 2022

Domain	n	ICT resources coefficient	SE	p
READ	5,982	17.76	2.36	$< .001$
MATH	5,982	18.17	2.48	$< .001$

ICTRES is standardised within system. Both outcomes use all ten plausible values, final weights, all 80 BRR replicates and Rubin combination; complete cases on ESCS, gender, grade, repetition and ICTRES. Reading is associated with ICT resources and is not a technology-unrelated negative control. Coefficients are achievement points per predictor SD.

Table A14: School-grouped out-of-fold regression calibration, Viet Nam, 2022

Outcome	n	Schools	R^2	RMSE	Intercept	Slope
Mathematics	6,068	178	0.371	68.023	17.494	0.962
Anxiety	5,934	178	0.199	0.851	0.054	0.877

Final student weights; five-fold GroupKFold keeps each school in one fold; LightGBM handles missing predictors natively and missing targets are excluded. Mathematics results average ten PV-specific diagnostics; anxiety uses the observed index. Intercept/slope come from weighted observed-on-predicted regressions (ideal 0/1), without recalibrating the validation predictions. Intercepts and RMSE use the outcome units. These diagnostics are descriptive, not design-based confidence estimates.

Table A15: Analytic samples in distraction and digital-habit figures, PISA 2022

System	M1	M2	A1	A2	M4
BRN	2,461	2,446	2,273	2,237	774
HKG	2,643	2,633	2,628	2,608	908
IDN	6,078	6,020	–	–	2,476
JPN	2,672	2,657	2,652	2,616	1,122
KHM	4,503	4,493	4,014	3,887	–
KOR	2,947	2,939	2,928	2,917	1,238
MAC	2,075	2,068	2,058	2,049	716
MYS	3,055	3,019	2,923	2,861	876
PHL	3,180	3,138	–	–	1,341
SGP	3,128	3,120	3,118	3,105	1,426
TAP	2,691	2,680	2,681	2,667	977
THA	3,903	3,864	–	–	1,485
VNM	5,913	5,912	5,811	5,723	–

Unweighted complete-case student counts. M1/M2/M4 model mathematics with ten plausible values; A1/A2 model anxiety. M4 includes digital habits in the computer-based systems. Dashes indicate no model for the corresponding outcome. These counts accompany the final-weighted BRR estimates.

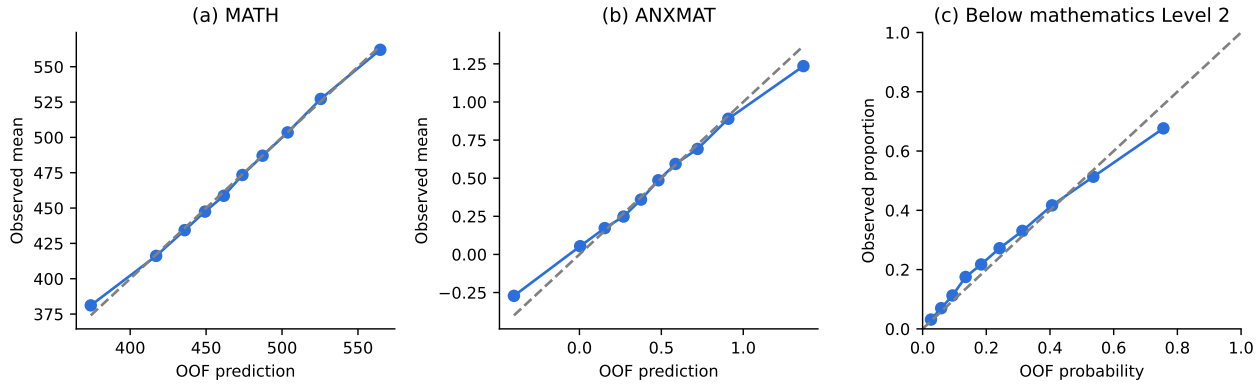


Figure A4: Out-of-fold reliability, Viet Nam, PISA 2022: (a) mathematics regression ($n = 6,068$, 178 schools); (b) anxiety regression ($n = 5,934$, 178 schools); (c) separate classification of mathematics below Level 2, threshold 420.07 points (OECD, 2023c). Points are final-weighted means within predicted deciles; the dashed diagonal indicates perfect calibration. Mathematics panels average all ten plausible values. No confidence bands or significance tests are implied.

The separate below-Level-2 classifiers have mean weighted Brier score 0.164, compared with 0.203 for each fold's training-prevalence baseline. Brier evaluates probabilities for this binary diagnostic, not the continuous regression. The predictions are retrospective associations using concurrent questionnaire data, not a validated prospective screening tool.

Table A16: Mathematics 2012–2022: link-error sensitivity in comparison systems

System	Difference	Sampling/PV SE	Link error	Total SE	95% interval
HKG	-20.9	4.4	3.6	5.7	[-32.0, -9.8]
IDN	-9.6	4.7	3.6	5.9	[-21.1, 1.9]
JPN	-0.8	4.6	3.6	5.9	[-12.3, 10.6]
KOR	-26.5	6.0	3.6	7.0	[-40.1, -12.8]
MAC	13.8	1.5	3.6	3.9	[6.2, 21.4]
MYS	-11.8	4.0	3.6	5.4	[-22.3, -1.3]
SGP	1.2	1.8	3.6	4.0	[-6.7, 9.1]
TAP	-12.7	5.0	3.6	6.2	[-24.8, -0.7]
THA	-32.8	4.4	3.6	5.6	[-43.9, -21.7]
VNM	–	–	–	–	–

Student final weights; cycle means combine BRR sampling and plausible-value uncertainty. Total SE is $\sqrt{SE_{2012}^2 + SE_{2022}^2 + 3.58^2}$, using the mathematics link error reported in NCES technical notes, Table 3 (sourced to OECD) (National Center for Education Statistics, 2023). This is a two-cycle difference, not an all-cycle trend model. Viet Nam is excluded because OECD does not establish a reliable link with its prior performance results (OECD, 2023b); adding generic link error would not repair that problem. The adjustment does not apply to questionnaire proportions or within-cycle regressions.